Teleoperation: Self-Driving Car Project

Capstone - ECE4773

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Team Introductions

Lili Nguyen (PM)

Van Lang (Software)

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Pundit Vorakitolan (Hardware)

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MiLab Introduction

MiLa
Mobility Intelligence Lab

Research lab led by Dr. Bin Xu, research assistant Professor at the University of Oklahoma

Focuses on intelligent mobility systems and data-driven vehicle technologies.

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Problem Statement: MiLab lacks a reliable long-distance communication system to remotely control a DBW-enabled car using a driving simulator.

Who it Affects ?	The Pain	What To Do ?
MiLab researchers & students	Unreliable long-distance simulator-to-DBW control.	Develop low-latency simulator-to-vehicle control for remote testing.

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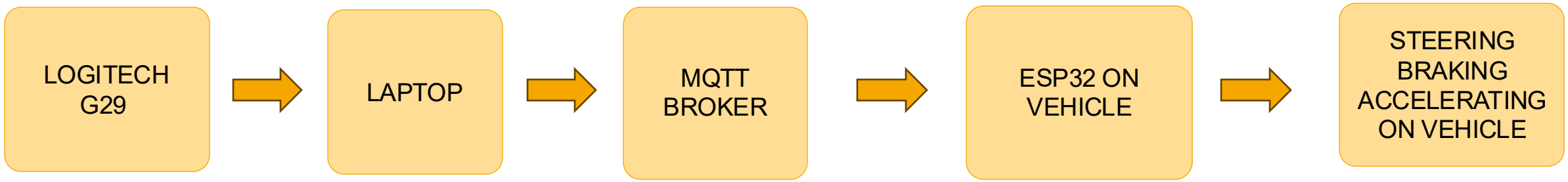
System Overview



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Quantitative Design Requirements

- Input Precision for Bluepad32
  - Requirement: The system must map 16 bit analog inputs from the Logitech G29
  - Metric: 256 discrete steps for the throttle/brake and steering (0 to 255)
- Data Throughput
  - Requirement: Maintain consistent telemetry stream
  - Metric: Update frequency at 100Hz to ensure smooth control loops

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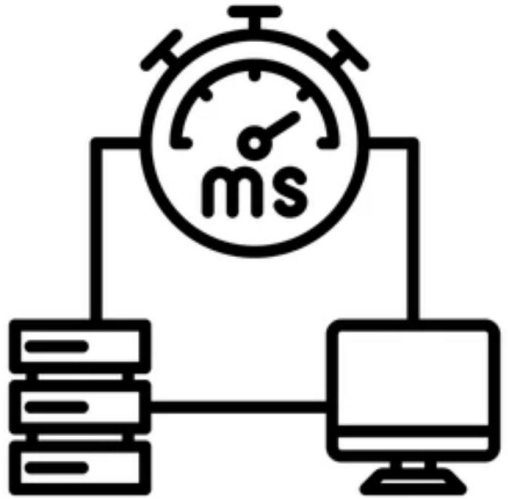
Performance Objectives

- Latency
  - Objective: minimize the latency from the driving simulator
  - Target: Latency < 200ms
- Connection Stability
  - Objective: Maintain connection without drops
  - Target: Successful packet rates during the driving session



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Source Code

- Bluepad32 uses a virtual gamepad
- Uni_gamepad.h
  - Data structure for steering, throttle, and braking
- Uni_controller.h
  - Manages bluetooth and connection handshakes
- Take these principles from Bluetooth and apply them to our system

Technical Approach

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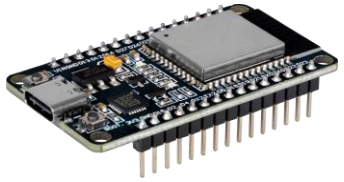
serial

wireless



- Get a working MVP, then rapidly iterate on it
  - MVP will likely be similar to the other two capstone teams
    - We all have the same hardware so no Bill of Materials
  - Ultimate goal is to reduce latency; we want to test as many differing ideas as possible
  - Allows for course correction and evolving to help the lab where it needs it the most
- MVP approach
  - Interface between a Logitech G29 and an ESP32 using python
  - Use the lab's incumbent protocol (MQTT) to transmit messages via intermediary (broker)
  - Once messages arrive at the car ESP32/relay/DAC PCB board, hand-off to DBW team
- Improving the MVP
  - Send telemetry/feedback data from the car
  - Timestamp and track down/fix root cause of latency

Focus: use software to wirelessly interface between a driving simulator and a car. Iterate on that MVP, optimize latency, and control the car well enough to drift.

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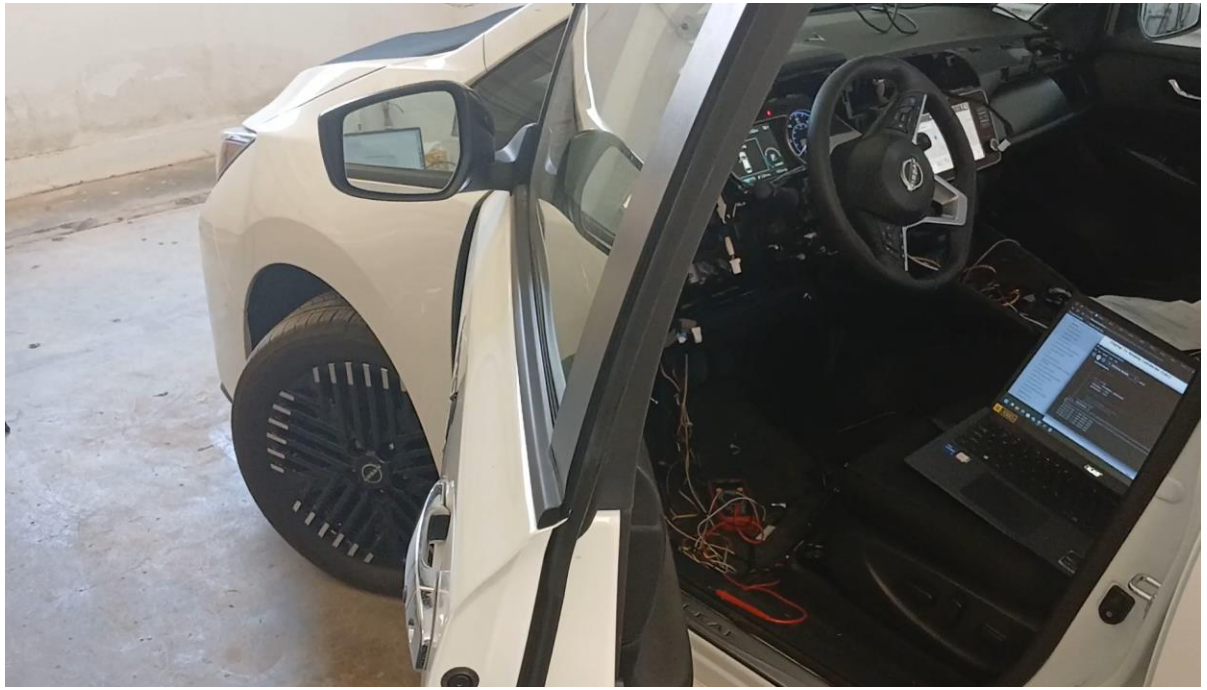
Technical Approach

- We have a DBW (drive-by-wire) system to integrate our system into
  - We also have a PCB with an ESP32/relays/DAC on the car side; we also want to design a smaller board to house those components
- Our main technical problem to solve after building the MVP is the latency between device and car, illustrated by the Bluetooth solution



Our driving sim/ESP32 duo integrates into this system

Latency – it's a problem!

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Latency & Upgrades

- A PS5 controller over bluepad32 should only add a few milliseconds
  - Our 1-2 sec latency comes from another source
  - We should look at timestamp logging to identify the issue
  - Adding another network layer will only exacerbate the problem
  - Slow ESP32/car ECU processing likely suspect?
- Start with pygame and upgrade to psydl2 when we need to optimize latency
- Add features like force feedback on the wheel

Latency – it's a problem!

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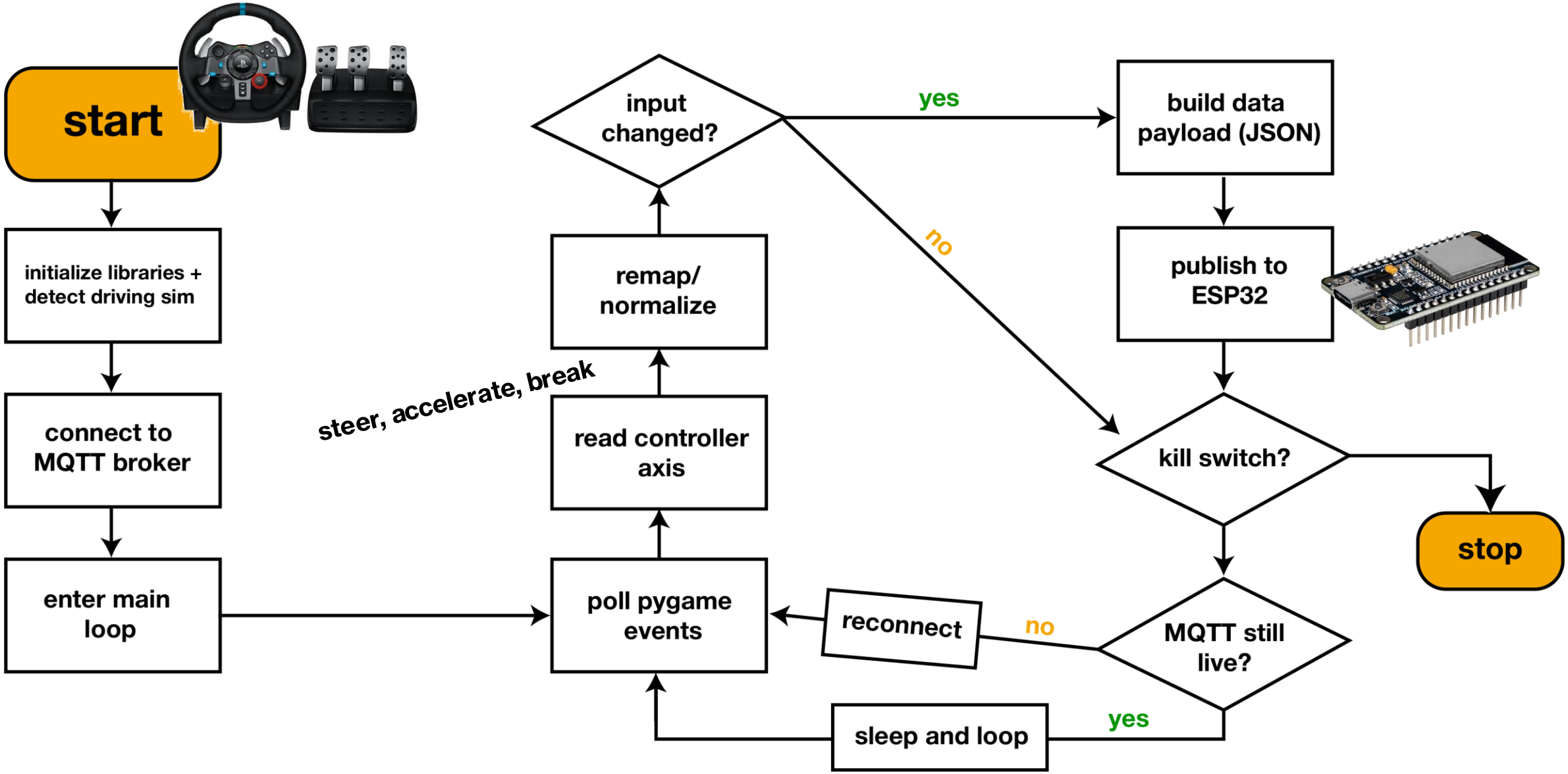
Driving Sim Software Block Diagram



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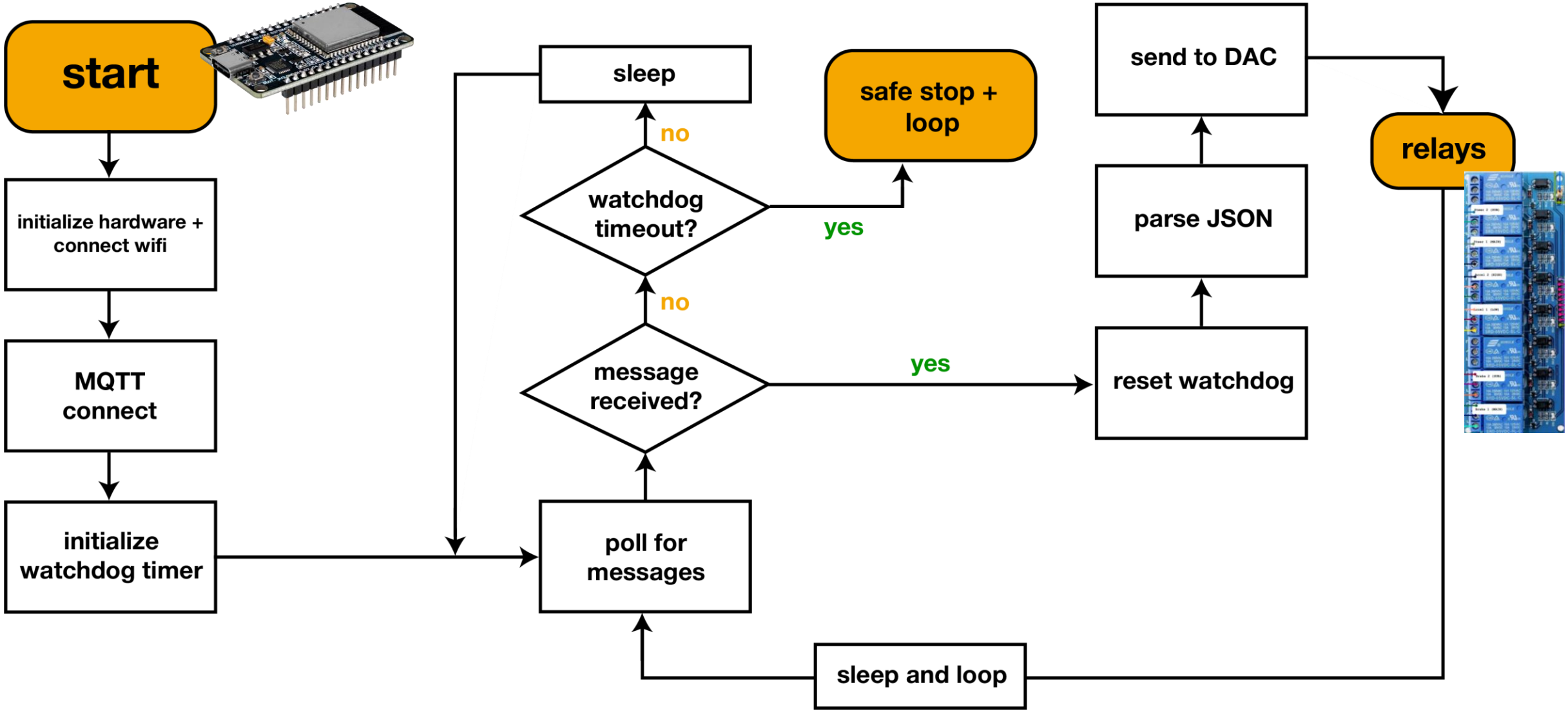
ESP32 Software Block Diagram



ESP

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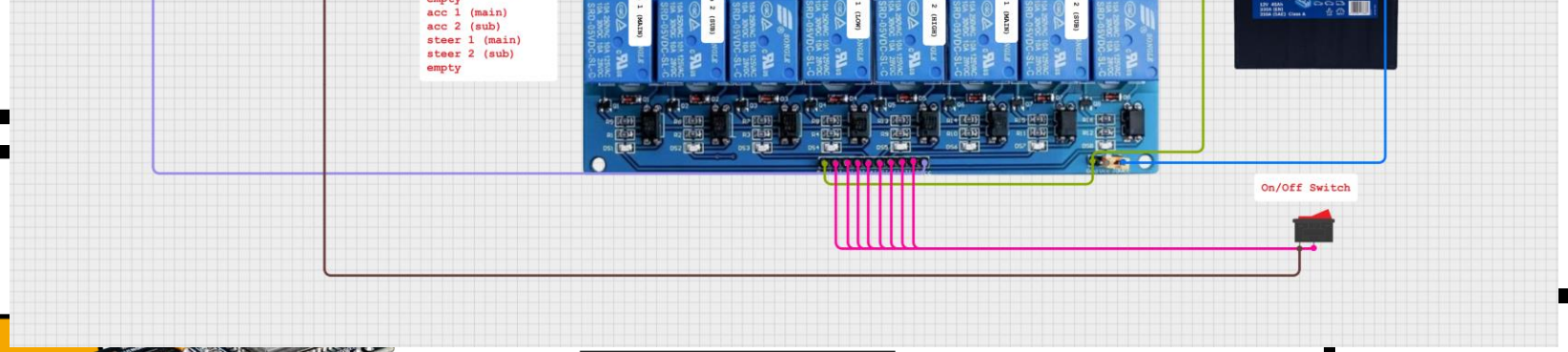
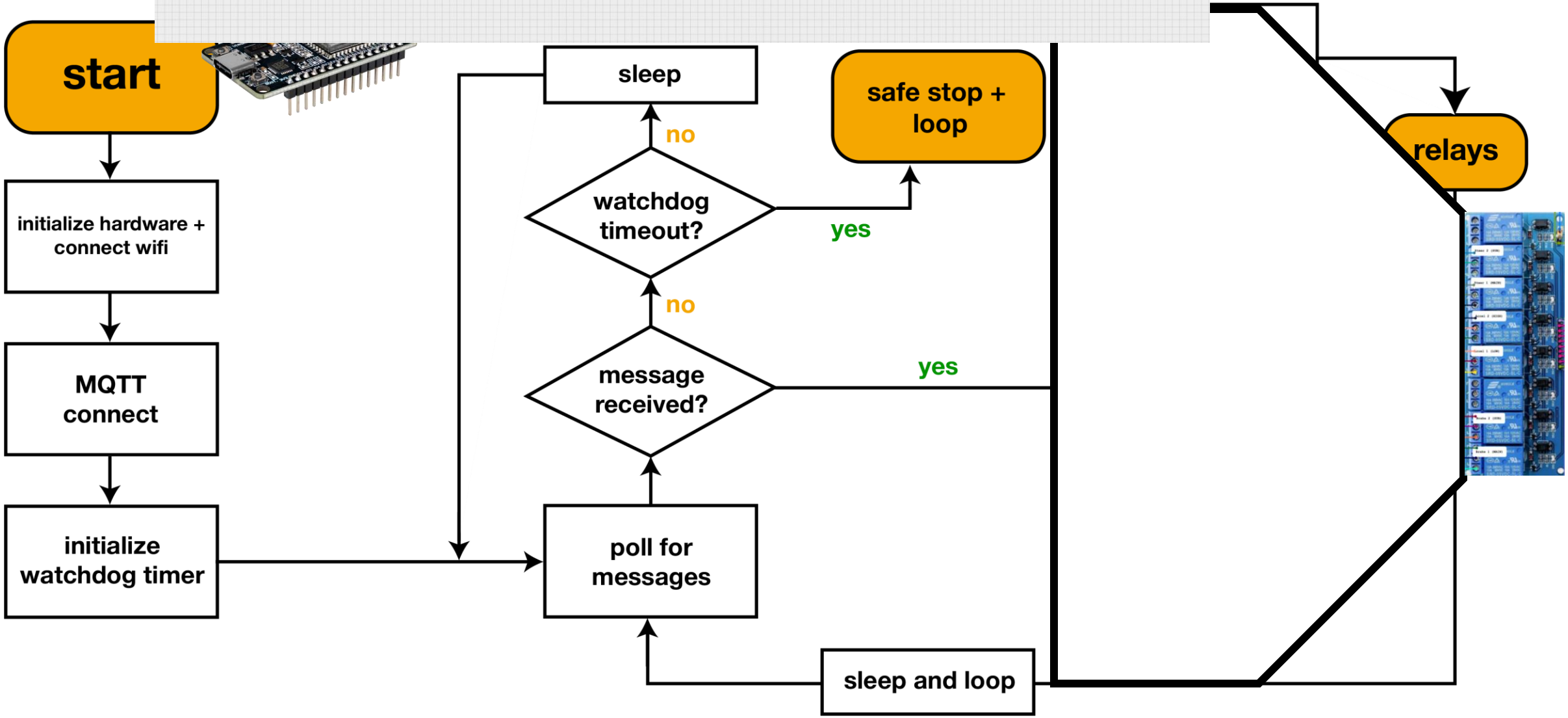

acc 1 (main)
acc 2 (sub)
steer 1 (main)
steer 2 (sub)
empty
On/Off Switch


start
initialize hardware + connect wifi
MQTT connect
initialize watchdog timer
sleep
no
watchdog timeout?
yes
safe stop + loop
no
message received?
yes
poll for messages
sleep and loop
relays

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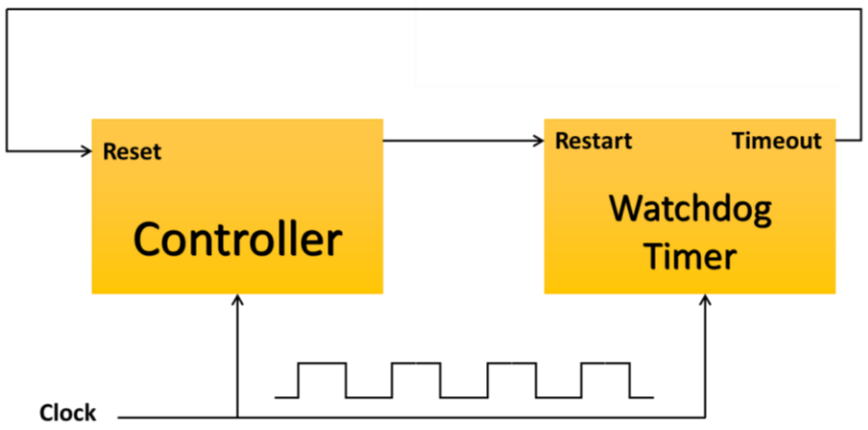
Safety Considerations

- Controlled Testing
- Hard Stop
- Operator Awareness
- Comms Loss Behavior



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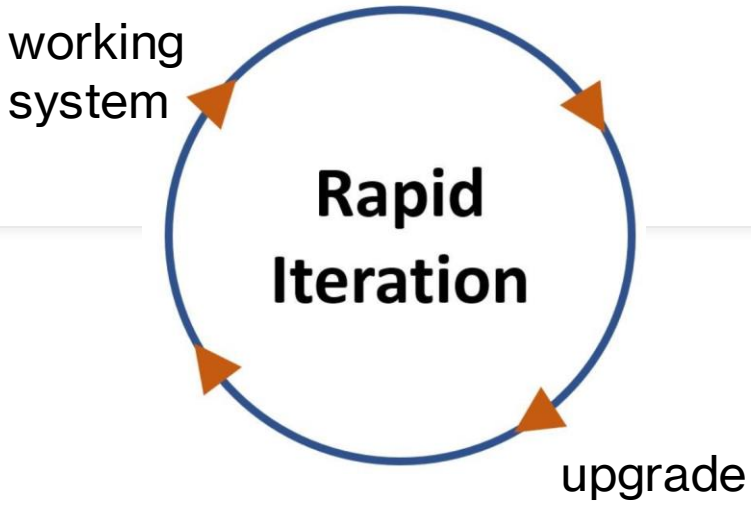
Major Tradeoff Issues



- Simplicity -> Complexity
- MQTT vs other protocols
  - Other protocols/methods of remotely communication may reduce latency
  - MQTT is actively being used by this lab
  - Better to start simple and add complexity later
- G29 vs PS5 controller
  - The G29 is more reliable and provides a closer simulation to real life
  - Instead of Bluetooth, we can use a more customizable communication protocol

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Cost/Budget

- Main costs are small parts:
  - ESP32 boards
  - wires, connectors, adapters
  - power parts (regulator, fuse, switch)
- Cost control
  - Use lab's existing DBW system, share the same expensive hardware with other teams
- Budget risk: extra parts if hardware breaks
- Plan: pre-test early to avoid damage and reduce rework

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Milestone Chart/Schedule

1

Preliminary Design Phase (2/08–2/22)

- Define system architecture and requirements
- Draft interfaces, message schema, and initial prototype plan

2

Critical Design Phase (2/22–4/01)

- Implement and integrate core system (end-to-end path)
- Add safety behavior (timeouts, E-stop) and validation tests

3

Final Demonstration Phase (4/01–4/22)

- Plan to run stable demo, measure latency
- Execute final test checklist and capture results
- **Prepare Project Repository**
- **Design Archives**
- **Written Report**

Q&A